# Oluwatosin Ayeni

ML Engineer, LLM Training & Inference Optimization — Nebius

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## PROFILE

ML Engineer with production experience in LLM training, fine-tuning, and inference optimisation across H100 NVL and RTX 4090 hardware. Benchmarked vLLM v0.30 vs SGLang v0.5.20 on 2× H100 NVL — 1,443× TTFT speedup via RadixAttention, 5,965 tok/s peak, 100% SLO compliance at concurrency 128. Full LLM lifecycle: LoRA/PEFT fine-tuning (Mistral 7B, Phi-4, Qwen3), trained LM from scratch, AWQ/GPTQ quantisation, CUDA KV cache profiling. Open to London, Amsterdam, or remote-Europe.

## CORE COMPETENCIES

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| **LLM Inference & Training** | vLLM, SGLang, TensorRT-LLM, ONNX Runtime; PyTorch, PEFT/LoRA, Transformers; BFloat16/FP8 |
| **GPU & CUDA Optimisation** | NVIDIA Nsight Systems/Compute, CUDA kernel profiling, memory bandwidth analysis, warp occupancy, KV cache ops |
| **Inference Research** | Speculative decoding, continuous batching, RadixAttention/prefix caching, AWQ/GPTQ quantisation, flash attention, tensor parallelism |
| **LLM Architecture** | MHA/GQA/MLA, Dense vs MoE, VRAM modelling, tokeniser training, Qwen3, Mistral, Phi-4, Gemma 2 |
| **Infrastructure & Serving** | Kubernetes (HPA, GPU pools), Docker, Prometheus, Grafana, FastAPI, Python, GitHub Actions |

## PROFESSIONAL EXPERIENCE

Jan 2025–present — **Founder & Lead Engineer** — *Nexora Academic*, Glasgow, UK

- H100 NVL benchmark: vLLM v0.30 vs SGLang v0.5.20 — Qwen3-8B BF16, TP=2. Peak: 5,965 tok/s; RadixAttention: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context; 100% SLO compliance up to concurrency 128. Live: inference-web-theta.vercel.app
- Built *inference-optimizer*: production AI Inference Gateway — 3.87× faster (parallel execution), 61.4% cost reduction (prompt caching), 14× throughput (concurrency), 64.3% cheaper (model routing). 13 optimisations, 4 providers.
- Built *vllm-k8s-stack*: Kubernetes LLM serving — KV-cache-aware routing, HPA autoscaling on GPU pools, Prometheus/Grafana observability.
- Built *inference-harness*: unified benchmark framework spanning closed APIs + vLLM/SGLang/TensorRT-LLM with GPU profiling and live dashboard.
- Designed 13-phase AI Inference Engineering curriculum covering GPU architecture, CUDA optimisation, serving frameworks, and distributed inference.

Jan 2023–present — **Founder & AI Consultant** — *Vertinab / AnaburyAI Technology*, Glasgow, UK

- 28× LLM cost reduction deploying Qwen2.5-7B on RTX 4090 via vLLM/SGLang with continuous batching and KV cache optimisation.
- CUDA KV cache kernel profiling with NVIDIA Nsight: identified memory bandwidth bottleneck; 3.8× throughput improvement; 18× latency reduction via continuous batching redesign.
- Fine-tuned Mistral 7B with LoRA/PEFT on radiology corpus: +22% diagnostic accuracy, −35% hallucination rate; deployed as production clinical inference API.

2024–present — **Senior Software Engineer (Contract)** — *EduTechConnect / Stellar Labs*, Remote

- Real-time NLP pipeline (WhatsApp → Zoho CRM → Supabase) with sub-500 ms end-to-end latency under concurrent load.

Mar 2026–present **AI Automation Engineer** — *EduTech Global*, Remote

- LLM-backed student chatbots with structured output extraction and multi-step reasoning for Ahmadu Bello and Babcock universities.

2018–2022 **Data Analyst** — *Bytelab Technologies Ltd*, Nigeria

## KEY PROJECTS

**llm-inference-benchmark (H100)** vLLM v0.30 vs SGLang v0.5.20 on 2× H100 NVL 94 GB, Qwen3-8B BF16, TP=2. RadixAttention: 1,443× TTFT speedup. Peak 5,965 tok/s. SLO 100% at concurrency 128. Live results

**inference-harness** Unified benchmark harness spanning closed APIs + vLLM/SGLang/TensorRT-LLM with GPU profiling and live dashboard. GitHub

**phase3-model-architecture** Model architecture labs: transformer internals, GQA/MQA/MLA, Dense vs MoE, VRAM modelling, speculative decoding. GitHub

**Qwen3-TinyStories** LM trained from scratch: tokeniser training, architecture config, training loop, generation evaluation. GitHub

**My_finetuned_Phi4** Phi-4 LoRA fine-tuning on Cosmopedia: full pipeline — data prep, training, evaluation, inference. GitHub

## EDUCATION

2022–2023 **MSc Artificial Intelligence and Applications, Distinction** — *University of Strathclyde*, Glasgow, UK

2010–2016 **MTech / BSc Physics** — *Federal University of Technology Akure*, Nigeria

## OPEN SOURCE

**GitHub** 43 public repos — github.com/AyeniOluwatosinOlawale